

Examiner reports

Q1.

A lot of the candidates found this question very difficult. There were a number of good clear force diagrams, but there were many that included a friction force. Some also introduced a force acting down the slope. A few candidates did not make the directions of the forces clear. Many candidates were unable to resolve horizontally in part (b). The most common error was to resolve the 8 N force instead of the reaction force, obtaining equations such as $R = 8 \cos 30^\circ$. In part (c), many candidates produced results such as $R = mg \cos \theta$, by resolving perpendicular to the slope and ignoring the 8 N force. A few simply stated that $R = mg$.

Q2.

Part (a) was usually done very well. However there were very few good responses to part (b). The most common error was to try to find the magnitude of the resultant of P and the friction, assuming that the friction force had magnitude 58.8 N. For many candidates their understanding of the friction inequality seemed poor. In part (c), the candidates tended to either find the correct value or to simply give the answer 16 N, which was the product of the mass and the acceleration. Many candidates did part (d) very well, but a few omitted the negative sign from the acceleration. Those candidates who had difficulties usually used an incorrect acceleration.

Q3.

In part (a), there were a number of candidates who simply seemed to think that the acceleration was given by distance over time. Many of these candidates also gave the units as m s^{-1} , which further confirmed that their approach was incorrect. However their attempts at part (b) were very much better, probably helped by the printed answer. There were also a number of good solutions to part (c). The most common error in part (c) was to ignore the acceleration when calculating the friction force.

Q4.

In part (a) of this question, some candidates used an incorrect angle when trying to resolve. Some others simply assumed that the normal reaction force was equal to the weight. In part (b), those who had had difficulties resolving in part (a) had further difficulties here. However there were a reasonable number of candidates who were able to find the correct coefficient of friction. A large number of candidates did not include an inequality in their working. In part (c), a good number of candidates calculated the magnitude of the friction force correctly, but some then went on to assume that this was the only force acting.

Q5.

Again there were a good number of complete solutions to this question. Some candidates did have difficulties with parts (c) and (d). This was generally due to ignoring the friction force and simply assuming that the tension was equal to the product of the mass and the acceleration.

Q6.

This question was generally done well. Some candidates lost marks in part (a) because they ignored the instruction “By forming an equation of motion for each particle...”. A few candidates produced a negative acceleration in part (a), which was satisfactory because it was consistent with their equations, but then substituted a positive acceleration into an equation in part (b) to find an incorrect value for the tension.

There were many candidates who answered part (c) well including some of those who had had difficulty with the previous parts.

Q7.

The candidates found this question more difficult. Some treated it like a connected particles question and introduced an acceleration even though the system was in equilibrium. Candidates who persisted in using this approach made very little progress. While a number of candidates did produce complete solutions, there were a variety of errors, often involving resolving, that were made by a number of candidates.

Q8.

While there were a number of good solutions to this question, there were a number of difficulties that seemed to originate from not realizing that the reaction force at A acted downwards. This was a particular problem in parts (a) and (c). The printed answer seemed to help quite a lot of the candidates in part (b).

In part (d), the solutions tended to be either very good or very poor. Candidates who made a good start were normally able to find both reactions correctly. However, some hindered themselves by taking moments about points other than A and B or by taking moments, but not expressing these as part of an equation.

Q9.

A very small number of candidates seemed unsure how to deal with the vectors and made very little progress with the question. The majority of candidates did produce quite good solutions. Part (a) did not cause any particular difficulties. In part (b), some

candidates wrote $\sqrt{9^2 - 4^2}$ instead of $\sqrt{9^2 + 4^2}$. Those who could visualise part (c) had no difficulty, but some found it hard to identify a triangle to work in and tried to use the sine or cosine rules. A few candidates produced incorrect expressions involving sin or cos instead of using tan.

Q10.

There were many good responses to this question, with a good number of candidates gaining full marks. In part (a), the most common error was where candidates did not use a constant acceleration equation but used the argument $v = 0.8/2 = 0.4$, $a = 0.4/2 = 0.2$. In part (b), the most common error was due to incorrect signs in the equation of motion. In part (c), some candidates omitted a term from the equation and some did not give the answer to two significant figures as instructed.

Q11.

Drawing the force diagram in part (a) did cause difficulties for the candidates with several

wanting to include a force directed down the slope. Part (b) was done correctly by almost all of the candidates. In part (c), several candidates did not form a three term equation, either omitting the component of the weight down the slope or the friction force. Most candidates produced a good attempt at part (d).

Q12.

There were many good solutions to this question. There were a number of candidates who included additional forces on their force diagrams. Candidates should be encouraged to only show the forces acting on force diagrams. Including velocities or components as well as the forces often confuses diagrams and can cause marks to be lost. Part (b) was done very well. In part (c), the most common error was only to include one force, typically the friction force, and not to include the component of the weight that acts down the slope. Two other less common errors were to use $80ga$ instead of $80a$ and to resolve the weight incorrectly or not at all.

Q13.

There were some candidates who did not know how to approach the question. Some candidates produced an equation for vertical equilibrium, but did not take moments correctly, often not multiplying the reaction forces by a distance.

Q14.

There were many good responses to this question, with many candidates scoring full marks. Some candidates resolved horizontally and vertically, while others used a vector triangle approach combined with the sine rule. The most common error was to resolve incorrectly or use the wrong angles.

Q15.

Part (a) of this question was generally done very well, with many correct solutions. It was good to see a large number of candidates setting up an equation of motion for each particle and then using these to obtain the acceleration. The candidates seemed familiar with the topic and knew exactly what the question required.

The candidates found part (b) very much more demanding and there were relatively few correct solutions. Some candidates did not attempt this part of the question and some others had no clear strategy for finding a solution. One error was to find the time to travel 2m and then use this time to show that they travelled 2 m. Some candidates found the speed, when the rope became slack, but could not then progress to deal with the motion that took place after this had happened.

Q16.

Many of the candidates found this question more difficult than expected. Part (a) was generally done well, but problems then emerged in part (b). Here many candidates found the magnitude of the acceleration and then used this scalar quantity in part (c). It was not uncommon to see an equation that involved a velocity that was a vector and an acceleration that was a scalar. Some of the candidates who worked through correctly stopped in part (c) and did not find the distance that had been requested.

Q17.

There were a number of good responses to this question. Some parts caused the candidates a little more difficulty than the earlier questions. In part (a), some candidates included additional forces. Parts (b) and (c) were generally well done, the only significant problem was that some candidates used the wrong angle when finding the components of the weight and so ended up with their answers to parts (b) and (c) interchanged. This caused subsequent problems in part (d). However, most candidates made a good attempt at part (d). Part (e) was more challenging. Many candidates felt that the 14 kg would just slide down the slope. Some argued that the friction and weight would be proportional to the mass and so it would stay still. Quite a number repeated all the earlier calculations to obtain the same coefficient of friction and often concluded that the block would remain at rest.

Q18.

Part (a) was generally done very well. Very few candidates did not form two equations of motion as requested. A few had problems dealing with the friction force. Part (b) proved to be more difficult for some candidates, who dropped the negative sign and simply took the acceleration to be 1.225. Many of these candidates then just dropped a negative sign in the middle of their working. Part (c) was done well by some candidates, but again the negative acceleration caused problems for some.



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